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Measures of length

11.4 Measures of length

Old Testament

Name: Reed, rod (qaneh)

Equivalence: 6 cubits

Metric: 3 meters

Imperial: 10 feet

Name: Cubit (’amah)

Equivalence: —

Metric: 50 centimeters

Imperial: 18 inches

Name: Span (zereth)

Equivalence: 0.5 cubit

Metric: 23 centimeters

Imperial: 9 inches

Name: Palm (tefach)

Equivalence: 0.16 cubit

Metric: 8 centimeters

Imperial: 3 inches

Name: Finger (’etsba‘)

Equivalence: 0.04 cubit

Metric: 2 centimeters

Imperial: 0.75 inch

New Testament and Deuterocanon

Name: Schoinos

Equivalence: 30 stadia

Metric: 5.6 kilometers

Imperial: 3.5 miles

Name: Mile (milion)

Equivalence: 8 stadia

Metric: 1,480 meters

Imperial: 4,850 feet

Name: Sabbath day’s journey

Equivalence: 2,000 cubits

Metric: 1 kilometer

Imperial: 1,000 yards

Name: Stadium, furlong (stadion)

Equivalence: —

Metric: 185 meters

Imperial: 600 feet

Name: Fathom (orguia)

Equivalence: 0.1 stadium

Metric: 1.85 meters

Imperial: 6 feet

Name: Pēchus

Equivalence: 1 cubit

Metric: 50 centimeters

Imperial: 18 inches

Measures of length in the Old Testament

11.4.1 Measures of length in the Old Testament

Finger

11.4.1.1 Finger

References:

Hebrew אֶצְבַּע (ʼetsbaʿ) JER 52:21

JER 52:21: In this verse some common-language translations render the literal expression “four fingers” (RSV) as “three inches” (GNT), “seven centimeters” (SPCL), “eight centimeters” (GECLFRCLDUCL), or even “five centimeters” (Portuguese common language version). Older and more formal translations retain the “four fingers” of the original, and this is even done by some common-language versions (ITCLNOB). It seems likely that giving the thickness of the Temple columns as “four fingers” will communicate to the modern reader just as graphically as it did to the ancient one.

Palm

11.4.1.2 Palm

References:

Hebrew טֶפַח (tefach) 1KI 7:26, 2CH 4:5

Hebrew טֹפַח (tofach) EXO 25:25, EXO 37:12, EZK 40:5, EZK 40:43, EZK 43:13

Hebrew טַפְחָה (tafchah) PSA 39:6

The tefach /tofach is the width of the hand at the base of the fingers, not including the thumb.

EXO 25:25; EXO 37:12; 1KI 7:26; 2CH 4:5: In many languages it will be possible to express these words as the Hebrew does, as the width of a hand. Otherwise a measurement such as “8 centimeters” (FRCL) or “3 inches” (GNT) will do. For some reason CEV and NCV have “three inches” in the Exodus references but “four inches” in 1 Kings and 2 Chronicles.

EZK 40:5; EZK 43:13: Here the cubit is said to be longer by one tofach. This is the measure used throughout chapters 40–48 of Ezekiel. The shorter cubit actually measured about 45 centimeters (18 inches), and this longer one would have been 53 centimeters (21 inches). Some translations try to reflect the difference. Many, however, will prefer to use a rounded-off value of 50 centimeters (18 inches) for both.

PSA 39:6: Here the word tefach (tapchah) is a symbolic figure indicating the shortness of life. This may be the basis for the statement of Jesus in MAT 6:27. See the discussion under pēchus below.

Span

11.4.1.3 Span

References:

Hebrew זֶרֶת (zereth) EXO 28:16, EXO 28:16, EXO 39:9, EXO 39:9, 1SA 17:4, ISA 40:12, EZK 43:13

With the fingers spread, the zereth is the distance from the tip of the thumb to the tip of the little finger.

ISA 40:12: The picture here is of a person stretching out his fingers and using the distance between his thumb and little finger to measure the extent of the sky. For the second clause in this verse, GECL has “Who can … determine the extent of the sky with the span of his hand.”

Cubit

11.4.1.4 Cubit

References:

Hebrew אַמָּה (’amah) GEN 6:15, GEN 6:15, GEN 6:15, GEN 6:16, GEN 7:20, EXO 25:10, EXO 25:10, EXO 25:10, EXO 25:17, EXO 25:17, EXO 25:23, EXO 25:23, EXO 25:23, EXO 26:2, EXO 26:2, EXO 26:8, EXO 26:8, EXO 26:13, EXO 26:13, EXO 26:16, EXO 26:16, EXO 26:16, EXO 27:1, EXO 27:1, EXO 27:1, EXO 27:9, EXO 27:12, EXO 27:13, EXO 27:14, EXO 27:16, EXO 27:18, EXO 27:18, EXO 30:2, EXO 30:2, EXO 30:2, EXO 36:9, EXO 36:9, EXO 36:15, EXO 36:15, EXO 36:21, EXO 36:21, EXO 36:21, EXO 37:1, EXO 37:1, EXO 37:1, EXO 37:6, EXO 37:6, EXO 37:10, EXO 37:10, EXO 37:10, EXO 37:25, EXO 37:25, EXO 37:25, EXO 38:1, EXO 38:1, EXO 38:1, EXO 38:9, EXO 38:11, EXO 38:12, EXO 38:13, EXO 38:14, EXO 38:15, EXO 38:18, EXO 38:18, NUM 11:31, NUM 35:4, NUM 35:5, NUM 35:5, NUM 35:5, NUM 35:5, DEU 3:11, DEU 3:11, DEU 3:11, JOS 3:4, 1SA 17:4, 2SA 2:24, 2SA 8:1, 1KI 6:2, 1KI 6:2, 1KI 6:3, 1KI 6:3, 1KI 6:6, 1KI 6:6, 1KI 6:6, 1KI 6:10, 1KI 6:16, 1KI 6:17, 1KI 6:20, 1KI 6:20, 1KI 6:20, 1KI 6:23, 1KI 6:24, 1KI 6:24, 1KI 6:24, 1KI 6:25, 1KI 6:26, 1KI 7:2, 1KI 7:2, 1KI 7:2, 1KI 7:6, 1KI 7:6, 1KI 7:10, 1KI 7:10, 1KI 7:15, 1KI 7:15, 1KI 7:16, 1KI 7:16, 1KI 7:19, 1KI 7:23, 1KI 7:23, 1KI 7:23, 1KI 7:24, 1KI 7:27, 1KI 7:27, 1KI 7:27, 1KI 7:31, 1KI 7:31, 1KI 7:31, 1KI 7:32, 1KI 7:32, 1KI 7:35, 1KI 7:38, 2KI 14:13, 2KI 25:17, 2KI 25:17, 2KI 25:17, 1CH 11:23, 2CH 3:3, 2CH 3:3, 2CH 3:3, 2CH 3:4, 2CH 3:8, 2CH 3:8, 2CH 3:11, 2CH 3:11, 2CH 3:11, 2CH 3:12, 2CH 3:12, 2CH 3:13, 2CH 3:15, 2CH 3:15, 2CH 4:1, 2CH 4:1, 2CH 4:1, 2CH 4:2, 2CH 4:2, 2CH 4:2, 2CH 4:3, 2CH 6:13, 2CH 6:13, 2CH 6:13, 2CH 25:23, NEH 3:13, EST 5:14, EST 7:9, ISA 6:4, JER 51:13, JER 52:21, JER 52:21, JER 52:22, EZK 40:5, EZK 40:5, EZK 40:7, EZK 40:9, EZK 40:9, EZK 40:11, EZK 40:11, EZK 40:12, EZK 40:12, EZK 40:12, EZK 40:12, EZK 40:13, EZK 40:14, EZK 40:15, EZK 40:19, EZK 40:21, EZK 40:21, EZK 40:23, EZK 40:25, EZK 40:25, EZK 40:27, EZK 40:29, EZK 40:29, EZK 40:30, EZK 40:30, EZK 40:33, EZK 40:33, EZK 40:36, EZK 40:36, EZK 40:42, EZK 40:42, EZK 40:42, EZK 40:47, EZK 40:47, EZK 40:48, EZK 40:48, EZK 40:48, EZK 40:48, EZK 40:49, EZK 40:49, EZK 41:1, EZK 41:1, EZK 41:2, EZK 41:2, EZK 41:2, EZK 41:2, EZK 41:2, EZK 41:3, EZK 41:3, EZK 41:3, EZK 41:4, EZK 41:4, EZK 41:5, EZK 41:5, EZK 41:8, EZK 41:9, EZK 41:10, EZK 41:11, EZK 41:12, EZK 41:12, EZK 41:12, EZK 41:13, EZK 41:13, EZK 41:14, EZK 41:15, EZK 41:22, EZK 41:22, EZK 42:2, EZK 42:2, EZK 42:4, EZK 42:4, EZK 42:7, EZK 42:8, EZK 42:8, EZK 42:16, EZK 43:13, EZK 43:13, EZK 43:13, EZK 43:13, EZK 43:13, EZK 43:14, EZK 43:14, EZK 43:14, EZK 43:14, EZK 43:15, EZK 43:17, EZK 43:17, EZK 45:2, EZK 47:3, ZEC 5:2, ZEC 5:2

The ’amah is the distance from the tip of the elbow to the tip of the middle finger, that is, about 50 centimeters or 18 inches. There is also a longer cubit mentioned in Ezekiel (see tefach above, 11.4.1.2 Span<REALIA:11.4.1.2>).

EZK 41:8: The Hebrew text here speaks of an ’amah ’atsilah. The word ’atsilah is problematic, perhaps referring to a length, or perhaps having something to do with a foundation or a terrace. Translations generally treat it as

a length, something that qualifies the length of a cubit: “a long cubit” (so RSV). Others indicate this by giving a longer length for the rod than someone might expect, for example, “ten and one-half feet” (NCV). If it is felt to be significant that a non-standard cubit is being used here, then this latter solution will be unacceptable.

Gomed

11.4.1.5 Gomed

References:

Hebrew גֹּמֶד (gomed) JDG 3:16

JDG 3:16: This single appearance of the gomed gives us little to indicate its length. Many commentators and translations take it to be equal to a cubit. Since Ehud was able to hide the sword under his garment, it probably could not have been any longer than a cubit, and it may have been shorter. CEV translates the Hebrew word cherev (usually rendered “sword”) as “dagger,” thus indicating it was short, but leaves the phrase about its length untranslated. Another possible model is “a short ... sword” (GECL).

Reed, rod

11.4.1.6 Reed, rod

References:

Hebrew קָנֶה (qaneh) EZK 40:3, EZK 40:5, EZK 40:5, EZK 40:5, EZK 40:6, EZK 40:6, EZK 40:7, EZK 40:7, EZK 40:7, EZK 40:8, EZK 41:8, EZK 42:16, EZK 42:16, EZK 42:16, EZK 42:17, EZK 42:17, EZK 42:18, EZK 42:18, EZK 42:19, EZK 42:19

The qaneh in Ezekiel was a measuring rod used to measure length. It had a fixed length of six cubits and a palm, that is, about 3.5 meters (11.5 feet). See also 1.8.6 Measuring rod, measuring line<REALIA:1.8.6>, and “Reed”<FLORA:5.1.4> in Plants and Trees in the Bible.

Measures of length in the New Testament and the Deuterocanon

11.4.2 Measures of length in the New Testament and the Deuterocanon

Pēchus

11.4.2.1 Pēchus

References:

Greek πῆχυς (pēchus) MAT 6:27, LUK 12:25, JHN 21:8, REV 21:17, JDT 1:2, JDT 1:2, JDT 1:2, JDT 1:2, JDT 1:3, JDT 1:3, JDT 1:4, JDT 1:4, 2MA 13:5, 1ES 6:24, 1ES 6:24, S3Y 1:24

The pēchus is the New Testament equivalent of the cubit (’amah in Hebrew).

In most cases the translator will be able to avoid awkward fractions and decimal points for the pēchus by using the formula, two cubits equal one meter or one yard.

MAT 6:27; LUK 12:25: One problem of interpretation here is that the Greek word pēchus is normally used of distance rather than of time. However, it may also be used of time, and that meaning is much more satisfactory in the present context. Moreover, the Greek word rendered “span of life” by RSV is more frequently used as a measurement of time than height. Modern translations tend to accept the meaning of a measure of time. If span is used as a measurement of time, the use of the word pēchus to qualify it would mean a brief span, whereas if it is used of a person’s height, it would be a significant amount (about half a meter or 18 inches). NAB renders the last part of MAT 6:27 as “add a moment to his life-span,” and GECL has “lengthen his life only by one day.”

Other possibilities are “add a little time to his life,” “live a little longer,” and “make his life last even a little longer.” Some translators will indicate a specific amount of time, such as “even one day” or “even one hour.”

JHN 21:8: The distance given here is approximately 100 meters or 100 yards. Translators will want to follow something like CEV “only about a hundred yards from shore.”

REV 21:17: The text gives a measurement of 144 cubits, that is, twelve times twelve. This seems to be a symbolic number, and many translations—among them common-language translations such as GWNCVITCLGECL —choose to leave the number in the text. ITCL also adds the metric equivalent in the text, saying “one hundred forty-four cubits (seventy meters).”

GNT takes this measurement, 144 cubits (72 meters; 236 feet), to be the height of the wall (also NJBREBFRCL, Phps). It may, however, refer to its thickness (NIVBRCL). Some commentators make the point that, for a city that is 1,500 miles tall, a wall only 72 meters (236 feet) tall is so small that it seems foolish. But it is also pointed out that the wall, in this case, is not for the protection of the city (inasmuch as the gates stay open all the time) but for its demarcation. In any case, the notion of height is preferable to that of thickness. A translation should opt for one or the other; simply to say, as RSV and others say, “a hundred and forty-four cubits” (or, “72 meters” or “236 feet”), without indicating that this is the height, does not qualify as a translation. An alternative translation model for the first half of this verse is “He also determined the height [or, thickness] of the wall [or, fence]. It was 72 meters [or, 236 feet].” For this verse see also A Handbook on The Revelation to John, pages 306–307.

Fathom

11.4.2.2 Fathom

References:

Greek ὀργυιά (orguia) ACT 27:28, ACT 27:28

ACT 27:28; ACT 27:28: The orguia was traditionally the measure of a man’s outstretched arms from fingertip to fingertip. It was reckoned at approximately 6 feet or about 1.85 meters. It was used as a technical nautical term to measure the depth of water. Water depth was measured with rope with a weight on the end, and this measure possibly derives from the way of counting off the arm lengths of rope. The measurements mentioned in this verse are about 40 meters (130 feet) and about 30 meters (100 feet). Where a language has a unit of measure that is used specifically for water depth (like the English word “fathom”), it may be used here.

Stadium, furlong

11.4.2.3 Stadium, furlong

References:

Greek στάδιον (stadion) 2MA 11:5, 2MA 12:9, 2MA 12:10, 2MA 12:16, 2MA 12:17, 2MA 12:29

Greek στάδιος (stadios) MAT 14:24, LUK 24:13, JHN 6:19, JHN 11:18, REV 14:20, REV 21:16

MAT 14:24: The Greek says that the boat was “many stadia” from land. It is not necessary to express this in any particular unit of measurement. GNT “the boat was far out in the lake” is a good model. (Some Greek manuscripts have “in the middle of the lake” instead of “many stadia from the land.” In either case, neither reading gives a specific measurement; both merely state that the boat was now some distance from the shore, and therefore any phrase that conveys this information will be faithful to the intent of the text.)

LUK 24:13: The distance given here in the best manuscripts is “sixty stadia,” which is about 11 kilometers (7 miles). Some manuscripts have “one hundred sixty stadia,” but this reading cannot be correct. Other ways of expressing “sixty stadia” might be “two hour’s walk” (Leyden, Sranan, Timorese) and “two leagues” (Tzeltal, where “a league” is commonly explained as “one hour’s walk”).

JHN 6:19: English translations generally give "three or four miles" (RSV). Where the metric system is used, you would say "five or six kilometers" (FRCL).

JHN 11:18: In this verse translations generally have "about two miles" (RSV) or "almost two miles." Where the metric system is used, you would say "about three kilometers." It would also be possible to say "about a half hour walk."

REV 14:20; REV 21:16: The numbers in these verses, 1,600 stadia and 12,000 stadia respectively, are probably symbolic and, as such, are more important than the exact distance they might represent. For this reason some translations prefer to retain the numbers and explain the distances in a note or in parentheses in the text. So, for example, ITCL has "twelve thousand stadia (more than two thousand kilometers)," while NIV says "12,000 stadia in length" and adds this footnote: "That is, about 1,400 miles (about 2,200 kilometers)."

Sabbath day's journey

11.4.2.4 Sabbath day's journey

References:

Greek σάββατον ἔχον ὁδόν (sabbaton echon hodon) ACT 1:12

ACT 1:12: Jewish sources approximately contemporary with the New Testament period defined "a Sabbath day's journey" as 2,000 cubits, which is about one kilometer or 1,000 yards. Some translations express this as "about half a mile" (GNT) or "about a kilometer" (PV). ITCL renders it in terms of the amount of walking time, "a half hour's walk." SPCL expands it to "exactly that which the law permitted one to walk on the day of rest," although this may be felt to be too cumbersome. Also, this rendering still does not tell the reader what distance was involved.

Mile

11.4.2.5 Mile

References:

Greek μίλιον (milion) MAT 5:41

MAT 5:41: There are a number of ways in which the distances can be expressed here. Many translations have maintained the literal mile, even in situations where the metric system is employed. "One mile" was presumably the distance a Roman soldier could force a Jew to carry his equipment. What is important here is not the exact distance, but the relation between "one mile" and "two miles." While this relation will be clear even in settings where miles are unknown, many translations choose to say "one kilometer" and "two kilometers" simply to avoid introducing the unknown word, "mile."

Actually, it is possible to substitute any distance that takes about 15–20 minutes to walk, or even to express it in walking time; compare the Rincon Zapotec rendering for this verse: "If anyone forces you to carry his load for half an hour, go with this person for a whole hour." In some cultures it may be natural to express the "mile" in terms of its original meaning in Latin, which is "a thousand paces."

Schoinos

11.4.2.6 Schoinos

References:

Greek σχοῖνος (schoinos) 2MA 11:5

which seems to have read “five stadia,” which is much too closesuring rod. In Revelation its length is not given but to Jerusalem. Already in ancient times the text was emended toonly the lengths of the things it measures. See also read “five schoinos.” 1.8.6 Measuring rod, measuring line<REALIA:1.8.6>.

The schoinos was a distance equal to thirty stadia, about 5.6 kilometers (3.5 miles). There is a textual problem in 2MA 11:5, which seems to have read “five stadia,” which is much too close to Jerusalem. Already in ancient times the text was emended to read “five schoinos.”

Measuring rod

11.4.2.7 Measuring rod

References:

Greek κάλαμος (kalamos) REV 11:1, REV 21:15, REV 21:16

Like the qaneh (see above), the kalamos was a measuring rod. In Revelation its length is not given but only the lengths of the things it measures. See also 1.8.6 Measuring rod, measuring line<REALIA:1.8.6>.